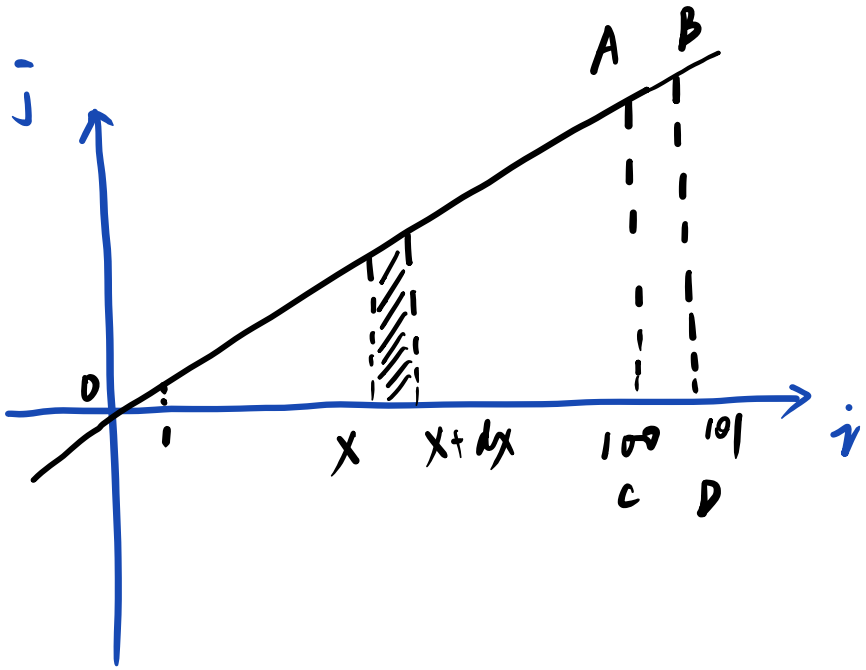
接续微分部分未完内容

未命名

$$\sum_{i=1}^{6} \frac{1}{i^2} = \frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{36}$$

希腊字母 sigma 的大写

$\sum_{i=1}^{100} i$: 著名的高斯 (^_^)



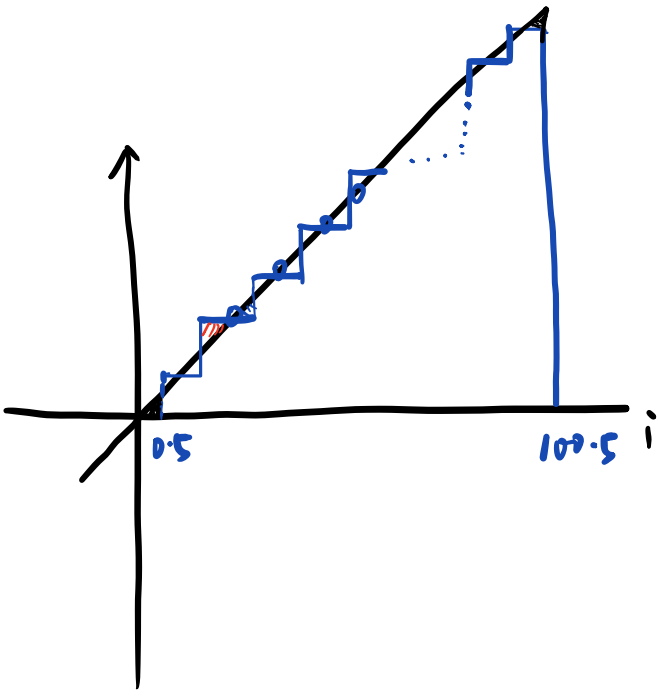
$$S_{OAC} = \int_0^{100} x\,dx = \frac{1}{2} \times 100^2$$

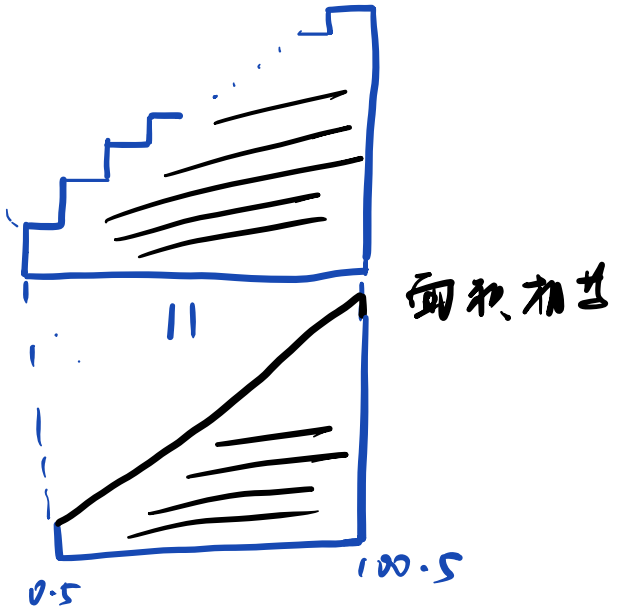
$$= \frac{1}{2}x^2 \Big|_0^{100}$$

$$\sum_{i=1}^{100} i = \sum_{i=1}^{100} i \times 1$$

① $= \int_1^{101} x\,dx$ ② $= \int_0^{100} x\,dx$

$= \frac{1}{2}x^2 \Big|_1^{101}$ (X) $= \frac{1}{2} \times 100^2$ (X)





做对并不容易！

微分，面积元，极限

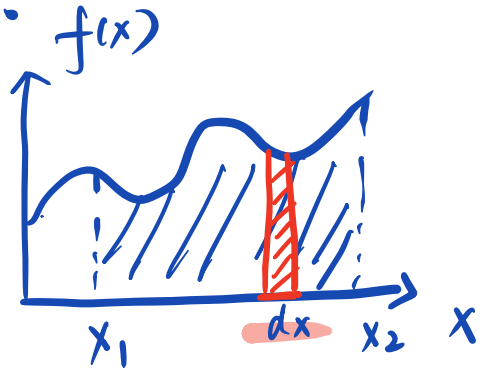
Answer：

$\frac{1}{2} \times (100+1) \times 100$

$= \frac{1}{2} \times (100.5+0.5) \times (100.5-0.5)$

(如图)

→ 不规则面积，体积



面积元：

$ds = f(x)\,dx$

$S_{阴影} = \int_{x_1}^{x_2} ds$

思：

规则：求围成面积

1° 怎么微分

2° 面积元是什么